

TCS349: Responsible and Explainable AI

Unit 1, Lecture 4

Bias in AI Systems: Historical, Data, Sampling, Measurement and Label Bias



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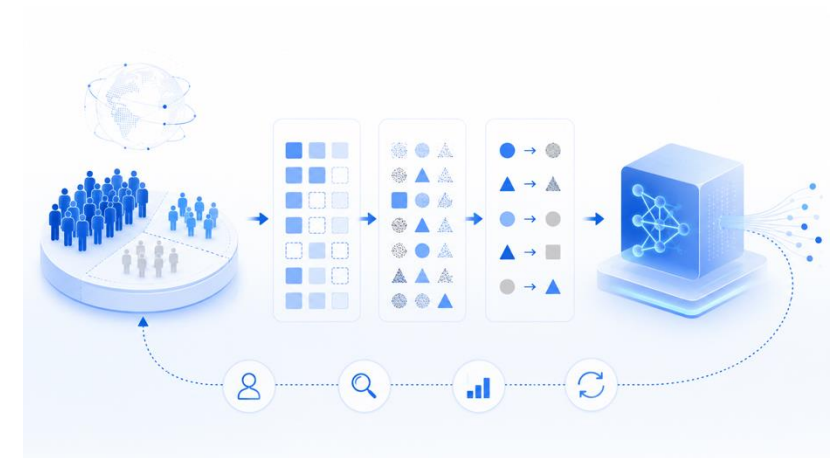
Warm-Up Question: Is all bias the same?

Suppose a model gives poor outcomes for one group of people.

Did the problem come from society itself, from the dataset, from the sample, from the way values were measured, or from the labels used in training?

Question for students: Can the same unfair result arise from different sources?

Key idea: To reduce bias responsibly, we must first identify where it entered the system.



What is Bias in AI?

In AI, bias refers to systematic distortion or unfairness that leads to unequal or misleading outcomes.

Bias does not always mean intentional discrimination.

It often arises from data, design assumptions, measurement choices, labelling practices or the social context itself.

An AI system can be technically functional and still be biased.

Why does Bias Matter?

Biased AI can deny opportunities, reinforce stereotypes and reduce trust.

It can affect hiring, lending, healthcare, education, policing and many other areas.

Bias may harm some groups more than others, even when the system seems efficient overall.

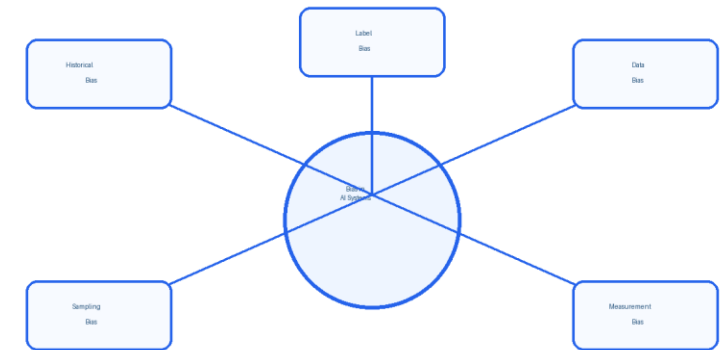
Responsible AI requires us to identify bias as early as possible in the pipeline.



Five Bias Types We Study Today

- Historical bias
- Data bias
- Sampling bias
- Measurement bias
- Label bias

These often interact with each other, so one AI problem may involve more than one bias type at the same time.



1. Historical Bias

Historical bias exists when the real world itself already contains inequality or unfair patterns.

Even if data is collected correctly, it may still reflect past social discrimination.

When AI learns from that history, it may reproduce the same injustice.

Example: past hiring or promotion decisions that favoured one gender more than another.



How to Recognise Historical Bias

Teaching Explanation

Ask: Does the dataset reflect a society that was already unequal?

Ask: Are some groups systematically underrepresented or disadvantaged in the real world?

Ask: If we copy the past, will we also copy the unfairness of the past?

Illustrative Example

Suppose a company historically hired very few women for technical roles.

A model trained on past hiring decisions may learn that male applicants are more 'suitable'.

The model may appear predictive but is ethically problematic.

2. Data Bias

Data bias arises when the dataset itself is incomplete, imbalanced or not sufficiently representative.

Important information may be missing, or some groups may appear too rarely.

The model then learns an unbalanced picture of reality.

Example: A face dataset that contains many examples of lighter skin tones but far fewer of darker skin tones.



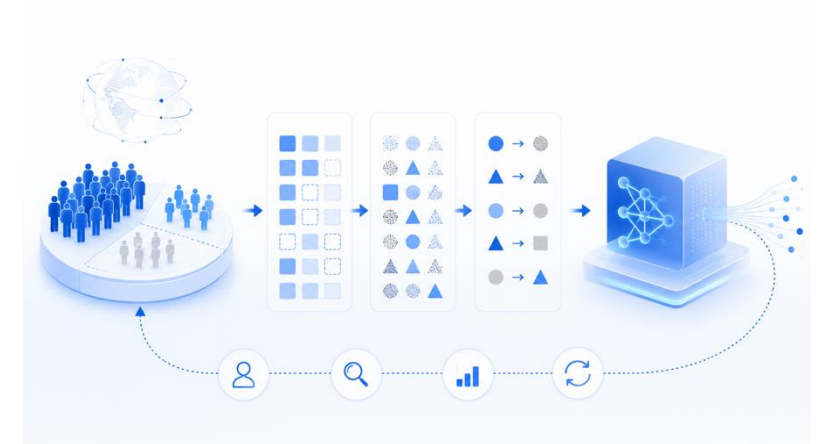
3. Sampling Bias

Sampling bias occurs when the method of selecting data is itself biased.

The collected sample does not properly represent the target population.

A model may perform well on the sampled group but poorly on others.

Example: Training a health app using data mainly from young urban users, then applying it to all age groups and regions.



4. Measurement Bias

Measurement bias appears when features, variables or proxies do not measure what we think they measure.

The quality of data collection tools, sensors or proxies may differ across groups.

This can create distorted inputs for the model.

Example: Using healthcare spending as a proxy for medical need, even though some groups may spend less because of unequal access.

5. Label Bias

Label bias occurs when the target labels used for training are themselves biased, inconsistent or subjective.

Labels are often created by humans, and human judgement may reflect stereotypes or institutional bias.

If the labels are problematic, the model learns a problematic target.

Example: If ‘good employee’ labels are based on biased manager evaluations, the model learns that biased judgement.

Quick Comparison of the Five Bias Types

Historical bias: unfairness already exists in society or context.

Data bias: the dataset content is skewed or incomplete.

Sampling bias: the selection of examples is unrepresentative.

Measurement bias: the variables or proxies are flawed.

Label bias: the target labels are biased or subjective.

Classroom Case: Facial Recognition System

Possible Bias Sources

Historical bias: broader social inequalities in surveillance practices

Data bias: too few examples of certain demographic groups

Sampling bias: images collected mainly from one region or population

Measurement bias: poor lighting or camera quality affects some faces more

Label bias: wrong identity labels during annotation

Teaching Discussion

Ask students: If the system performs poorly for one group, which bias type is most likely?

Then ask: Could more than one type be present together?

Final insight: biased outcomes often result from multiple interacting causes.